Department of Software Engineering Mehran University of Engineering and Technology
Shaheed Zulfiqar Ali Bhutto Campus, Khairpur Mir's

Course: SW 212 – Data Structure & Algorithms			
Instructor	Engr. Asmatullah Zubair	**PLOs**	1, 2, 3, 5, 12
Assign Date	25 October 2026	**Due Date**	11 November 2026
Batch	K25SW	**Assessment Score**	10 Mark

Open Ended Lab
Implement Data Structure in Inventory Management System

Course Learning Outcome

CLOs	Description	Domain	Taxonomy Level	PLO
1	Demonstrate various data structures such as arrays, lists, trees, graphs etc.	Cognitive	C3	1
2	Examine algorithms associated with each data structure.	Cognitive	C4	2
3	Perform the implementation of algorithms programmatically.	Psychomotor	P4	5
4	Comply with the rubrics to achieve the given tasks.	Affective	A2	8

Rubrics for Evaluation

	Components	Level of Achievement					
S. No.		Excellent (2)	Good (1.5)	Basic (1)	Just Acceptable (0.5)	Unacceptable (0)	Score (10)
1	Background Knowledge						
2	Select Appropriate Equipment and						
3	Analysis and Results						
4	Conclusions and Recommendations						
5	Report Writing						

Rubrics Description						
S. No.	**Components**	**Excellent (4)**	**Good (3)**	**Basic (2)**	**Just Acceptable (1)**	**Unacceptable (0)**
1	**Background knowledge**	Thorough study and all the questions have been answered correctly.	Adequate study and more than half of the questions have been answered correctly.	Sufficient study and half of the questions have been answered correctly.	Inadequate study and less than half of the questions have been answered correctly.	No study and none of the questions have been answered.
2	**Select Appropriate Equipment & Tools**	Relevant and smart selection of equipment and tools to achieve desired objective with proper reasoning.	Satisfactory selection of equipment and tools to achieve desired objective with proper reasoning.	Satisfactory selection of equipment and tools to achieve desired objective without proper reasoning.	Unsatisfactory selection of equipment and tools to achieve desired objective.	No use of equipment and tools leading towards major errors in the objective.
3	**Analysis & Results**	Appropriate data collected, correct analysis & results correctly interpreted.	Appropriate data collected but insufficient analysis & results adequately interpreted.	Inappropriate data collected but sufficient analysis & results inadequately interpreted.	Inappropriate data collected, no understanding of analysis & results incorrectly interpreted.	No data collected, analysis & results interpretation.
4	**Conclusions and Recommendations**	Significant findings are summarized. Precisely concluded. Excellent suggestion for further research.	Significant findings are summarized. Good conclusion. Good suggestion for further research.	Significant findings are summarized. Acceptable conclusion. Acceptable suggestion for further research.	Findings are poorly summarized. Poor conclusion. Poor suggestion for further research.	No findings are summarized. Poor conclusion. No suggestion for further research.
5	**Report Writing**	Report meets all requirements and it is prepared in original and creative way to engage readers.	Report meets all prescribed requirements.	The requirements of report writing are not properly addressed.	The report submitted but not according to requirements.	Report was not prepared or they have not required elements.

Task Objective

The goal of this task is to apply your knowledge of Data Structures and Algorithms (DSA) to design and implement a functional project that demonstrates the principles of efficient data organization and manipulation. This task will assess your ability to solve real-world problems using appropriate data structures and operations.

Task Description

You are required to design and develop a Data Structure Project that focuses on creating and managing an inventory system. The project must include the following key components and fulfill the specified requirements:

Requirements

1. Inventory Management System:

Develop a system to store and manage data for a specific inventory of your choice. Example inventories include:

- **Hospital Management System:** Patient records, doctor schedules, or medical supplies.
- **Grocery Store Inventory:** Product details, stock levels, or sales records.
- **Cricket Management System:** Player statistics, match schedules, or team data.
- **Student Database System:** Student profiles, grades, or attendance records.

You may choose any other inventory system relevant to your interests or group focus.

2. Implementation of CRUD Operations:

Your project must support the following basic operations for the inventory system using appropriate methods of the assigned data structure:

Create: Add new entries to the inventory (e.g., products, patients, players, or students).
Read: Display or retrieve data from the inventory.
Update: Modify existing records in the inventory.
Delete: Remove entries from the inventory.

You are encouraged to implement operations using functions such as:

Push() and **Pop()** for Stack-based operations.
Enqueue() and **Dequeue()** for Queue-based operations.
Add() and **Delete()** for Linked List operations.

3. Group Work and Data Structure Assignment:

The class will be divided into three groups, and each group will focus on implementing their inventory system using the assigned data structure:

Group A: Use a Linked List to manage the inventory.
Group B: Use a Stack to manage the inventory.
Group C: Use a Queue to manage the inventory.
Each group must utilize the core operations and properties of the assigned data structure to develop their project, ensuring that the structure's unique characteristics are evident in the implementation.

Deliverables

1. Upload a project report on Google Classroom in a word file that consists.
   - The chosen inventory system.
   - The implemented data structure code.
   - Output of how each CRUD operation works.
   - Explaining each operation.
2. Upload complete project on GitHub and attach link in project report.
3. Present project in a video explaining full project, upload video on G-Drive and attach link in project report.

Signature of Subject Teacher:  **Dated:25-10-2026**